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CODE A

Mock JEE Main – 7 (CBT) | JEE - 2024

Date: 14/03/2024 **Maximum Marks: 300**

Timing: 3:30 PM to 6:30 PM **Duration: 3.0 Hours**

General Instructions

1. The test is of **3 hours** duration and the maximum marks is **300.**
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices **(A)**, **(B)**, **(C)** and **(D)**, out of which **ONLY ONE CHOICE** is correct.
4. **Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 (any) questions have to be attempted. You will NOT be allowed to attempt the sixth question. If you wish to attempt any other question apart from the five already attempted, then you will have to delete any one response from the five previously answered and then proceed to answer the new one.**
   The answer to each question should be **rounded off to the nearest integer**.
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them**.

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.

Name of the Candidate (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ Invigilator's Signature _____

PART I: PHYSICS	MARKS: 100

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. An unstable heavy nucleus at rest breaks into two nuclei which move away with velocities in the ratio of 8: 27. The ratio of the radii of the nuclei (assumed to be spherical) is:

(A) 3 : 2 **(B)** 2 : 3 **(C)** 4 : 9 **(D)** 8 : 27

2. A particle of mass M at rest decays into two particles of masses m_1 and m_2 having nonzero velocities. The ratio of the de Broglie wavelengths of the particles, λ_1 / λ_2, is:

(A) m_1/m_2 **(B)** m_2/m_1 **(C)** 1.0 **(D)** $\sqrt{m_2}/\sqrt{m_1}$

3. Match the following:

List-1		List-2	
(A)	Coefficient of viscosity	**(P)**	$[M^2L^{-1}T^{-2}]$
(B)	Surface tension	**(Q)**	$[ML^0T^{-2}]$
(C)	Modulus of elasticity	**(R)**	$[ML^{-1}T^{-2}]$
(D)	Energy per unit volume of a fluid	**(S)**	$[ML^{-1}T^{-1}]$

(A) A-S, B-P, C-Q, D-R **(B)** A-P, B-Q, C-R, D-R

(C) A-S, B-Q, C-R, D-R **(D)** A-S, B-Q, C-R, D-P

4. A particle of mass m and charge q has an initial velocity $\vec{v} = v_0 j$. If an electric field $\vec{E} = E_0 \hat{i}$ and magnetic field $\vec{B} = B_0 \hat{i}$ act on the particle, its speed will double after a time.

(A) $\frac{\sqrt{3}mv_0}{qE_0}$ **(B)** $\frac{3mv_0}{qE_0}$ **(C)** $\frac{\sqrt{2}mv_0}{qE_0}$ **(D)** $\frac{2mv_0}{qE_0}$

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5. An eye specialist prescribes spectacles having combination of convex lens of focal length 40 cm in contact with a concave lens of focal length 25 cm. The power of this lens combination, in diopter, is:

(A) + 1.5 **(B)** –1.5 **(C)** + 6.67 **(D)** –6.67

6. The potential energy for a force field $\vec{F}$ is given by $U(x, y) = \cos(x + y)$. The force acting on a particle at position given by coordinates $(0, \pi/4)$ is:

(A) $-\frac{1}{\sqrt{2}}(\hat{i}+j)$ **(B)** $\frac{1}{\sqrt{2}}(\hat{i}+j)$

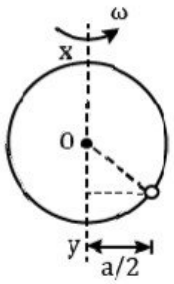
(C) $\left(\frac{1}{2}\hat{i}+\frac{\sqrt{3}}{2}j\right)$ **(D)** $\left(\frac{1}{2}\hat{i}-\frac{\sqrt{3}}{2}j\right)$

7. A spherical mirror forms an image of magnification 3. The object distance, if focal length of mirror is 24 cm, may be:

(A) 32 cm, 24 cm **(B)** 32 cm, 16 cm

(C) 32 cm only **(D)** 16 cm only

8. A small ring P is threaded on a smooth wire bent in the form of a circle of radius a and center O. The wire is rotating with constant angular speed ω about a vertical diameter XY, while the ring remains at rest relative to the wire at a distance $a/2$ from XY. Then ω^2 of is equal to:

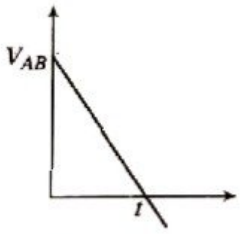


(A) $\frac{2g}{a}$ **(B)** $\frac{g}{2a}$ **(C)** $\frac{2g}{a\sqrt{3}}$ **(D)** $\frac{g\sqrt{3}}{2a}$

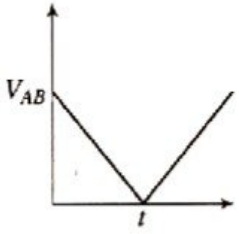
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9. A body A is thrown vertically upwards with such a velocity that it reaches a maximum height of h. Simultaneously, another body B is dropped from height h. It strikes the ground and does not rebound. The velocity of A relative to B versus time graph is best represented by (upward direction is positive)

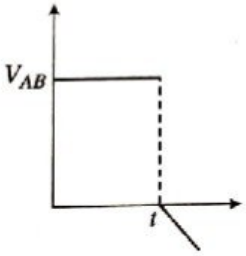
(A)



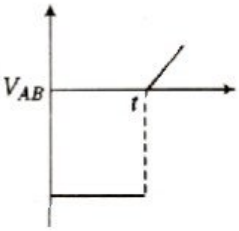
(B)



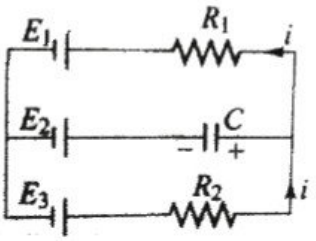
(C)



(D)



10. In the circuit shown (Figure), the batteries have emf $E_1 = E_2 = 1\ V$, $E_3 = 2.5\ V$, and the resistance $R_1 = 10\ \Omega$, $R_2 = 20\ \Omega$, Capacitance C = 10 μF. The charge on the left plate of the capacitor C at steady state is:



(A) +2 μC **(B)** –4 μC **(C)** –5 μC **(D)** +12 μC

11. A wire of length $l = 6 \pm 0.06$ cm and radius $r = 0.5 \pm 0.005$ cm has mass $m = 0.3 \pm 0.003$ g. Maximum percentage error in density is:

(A) 4 **(B)** 1 **(C)** 2 **(D)** 6.8

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12. A closed vessel contains 8 g of oxygen and 7 g of nitrogen. The total pressure is 10 atm at a given temperature. If now oxygen is absorbed by introducing a suitable absorbent, the pressure of the remaining gas in atm will be:

(A) 2 **(B)** 10 **(C)** 4 **(D)** 5

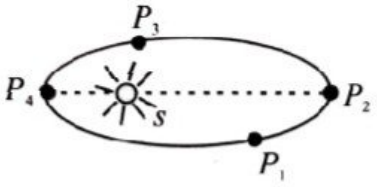
13. A person of mass 70 kg jumps from a stationary helicopter with the parachute open. As he falls through 50 m height, he gains a speed of 20 ms^{-1}. The work done by the viscous air drag is:

(A) 21000 J **(B)** –21000 J **(C)** –14000 J **(D)** 14000 J

14. In a region of space, the electric field is given by $\vec{E} = 8\hat{i} + 4j + 3k$. The electric flux through a surface of area 100 units in the *xy* plane is:

(A) 800 units **(B)** 300 units **(C)** 400 units **(D)** 1500 units

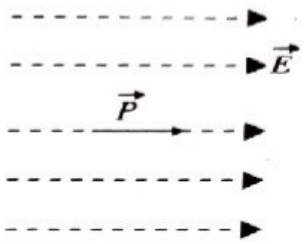
15. Figure shows a planet in an elliptical orbit around the Sun S. Where is the kinetic energy of the planet maximum?



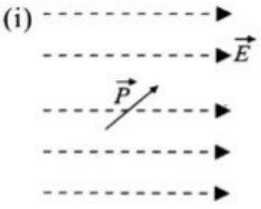
(A) P_1 **(B)** P_2 **(C)** P_3 **(D)** P_4

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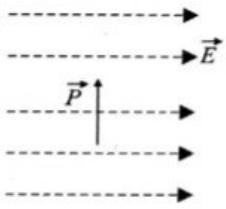
16. An electric dipole of dipole moment $\vec{p}$ is oriented parallel to a uniform electric field $\vec{E}$, as shown. It is rotated to one of the four orientations shown below. Rank the final orientations according to the change in the potential energy of the dipole-field system, most negative to most positive.



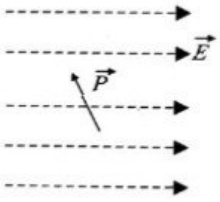
(i)



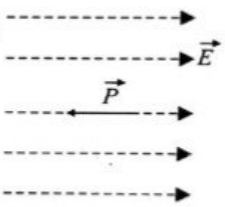
(ii)



(iii)



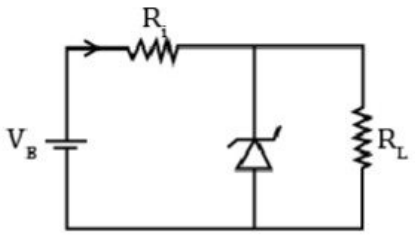
(iv)



(A) (i), (ii), (iv), (iii) **(B)** (iv), (iii), (ii), (i)

(C) (i), (ii), (iii), (iv) **(D)** (iii), (ii), (iv), (i)

17. The figure represents a voltage regulator circuit using a Zener diode. The breakdown voltage of the Zener diode is 6 V and the load resistance is, $R_L = 4k\Omega$. The series resistance of the circuit is $R_i = 1k\Omega$. If the battery voltage V_B varies from 8 V to 16 V, what are the minimum and maximum values of the current through Zener diode?



(A) 0.5 mA; 8.5 mA **(B)** 0.5 mA ; 6 mA

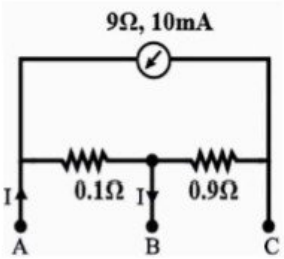
(C) 1.5 mA; 8.5 mA **(D)** 1 mA ; 8.5 mA

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18. A wire elongates by ℓ mm when a load w is hanged from it. If the wire goes over a pulley and two weights w each are hung at the two ends, the elongation of the wire will be (in mm):

(A) l **(B)** $2l$ **(C)** zero **(D)** $\frac{l}{2}$

19. A milliammeter of range 10 mA and resistance 9Ω is joined in a circuit as shown in figure. The meter gives full-scale deflection for current I when A and B are used as its terminals that is current enters at A and leaves at B (C is left isolated), the value of I is:



(A) 100 mA **(B)** 900 mA

(C) 1 A **(D)** 1.1 A

20. In a series LCR circuit the voltage across the resistance, capacitance and inductance is 10 V each. If the capacitance is short circuited, the voltage across the inductance will be :

(A) 10 V **(B)** $10/\sqrt{2}V$ **(C)** $(10/3)V$ **(D)** $20V$

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SECTION-2

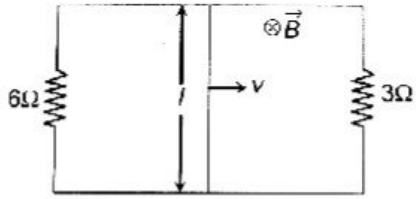
Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 (any) questions have to be attempted.

The answer to each question should be **rounded off to the nearest integer**.

1. A one metre long (both ends open) organ pipe is kept in a gas that has four times the density of air at STP. Assuming the speed of sound in air at STP is 300 m/s, the frequency difference between the fundamental and second harmonic of this pipe is _____ Hz.

2. An alternating voltage given by $V = 300\sqrt{2}$ sin (50t) (in volts) is connected across a $1\mu F$ capacitor through an AC ammeter. The reading of the ammeter (in mA) will be?

3. A rectangular loop with a sliding connector of length l = 1.0 m is situated in a uniform magnetic field B = 2T perpendicular to the plane of loop. Resistance of connector is $r = 2\Omega$. Two resistances of 6Ω and 3Ω are connected as shown in figure. The external force (in N) required to keep the connector moving with a constant velocity v = 2 m/s is?

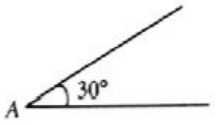


4. The trajectory of projectile, projected from the ground is given by $y = x - \frac{x^2}{20}$. Where x and y are measured in meter. The maximum height attained (in meter) by the projectile will be?

SPACE FOR ROUGH WORK

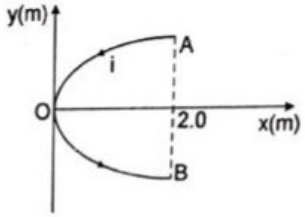
5. A solid sphere of mass 2 kg and radius 0.5 m is rolling with an initial speed of 1 m s^{-1} goes up an inclined plane which makes an angle of 30° with the horizontal plane, without slipping. If the sphere returns to the starting point A after $\eta \times 10^{-2}$ sec, then the value of η is?

(Take: g = 10 m/s^2)



6. M grams of steam at 100°C is mixed with 200 g of ice at its melting point in a thermally insulated container. If it produces liquid water at 40°C, then the value of *M* is _____ .[heat of vaporization of water is 540 cal/g and heat of fusion of ice is 80 cal/g]

7. A conducting wire bent in the form of a parabola $y^2 = 2x$ carries a current $i = 2\ A$ as shown in figure. This wire is placed in a uniform magnetic field $\vec{B} = -4k$ tesla. The magnetic force on the wire (in newton) is?



8. In a Young's double slit experiment, 12 fringes are observed to be formed in a certain segment of the screen when light of wavelength 600 nm is used. If the wavelength of light is changed to 400 nm, number of fringes observed in the same segment of the screen will be _____ ?

9. The water is filled upto height of 12 m in a tank having vertical sidewalls. A hole is made in one of the walls at a depth *h* below the water level. The value of h for which the emerging stream of water strikes the ground at the maximum range is _____ m.

10. A 1 m long copper wire carries a current of 1 A. If the cross section of the wire is 2.0 mm^2 and the resistivity of copper is $1.7 \times 10^{-8}\ \Omega m$, the force experienced by moving electron in the wire is _____ $\times 10^{-23}$ N. (charge on electron = 16×10^{-19} C)

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PART II: CHEMISTRY	MARKS: 100

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. For which of the following sets of quantum numbers, an electron will have the highest energy?

(A) 3, 2, 1, –1/2 **(B)** 4, 3, –1, –1/2 **(C)** 4, 1, –1, +1/2 **(D)** 5, 0, 0, –1/2

2. Which of the following statements regarding metal carbonyls is false?

(A) The metal carbon bond in metal carbonyl possesses both σ and π character.

(B) In $[Co_2(CO)_8]$, CO forms a bridge between metal atoms.

(C) The M-C π bond is formed by the donation of a pair of electrons from the filled bonding π orbital of CO into the vacant d orbital of metal.

(D) The metal to ligand bonding creates a synergic effect which strengthens the bond between CO and the metal.

3. Which of the following is correct?

(A) For the reaction:

$$PCl_5(g) \rightarrow PCl_3(g) + Cl_2(g) \quad ; \quad \Delta H = \Delta U$$

(B) For the reaction:

$$H_2(g) + Cl_2(g) \rightarrow 2HCl(g) \quad ; \quad \Delta H < \Delta U$$

(C) For the reaction:

$$FeCO_3(s) \rightarrow FeO(s) + CO_2(g) \quad ; \quad \Delta H > \Delta U$$

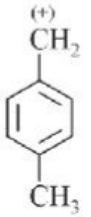
(D) For the reaction:

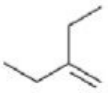
$$CO(g) + 3H_2(g) \rightarrow CH_4(g) + H_2O(g) \quad ; \quad \Delta H > \Delta U$$

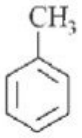
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4. Hyperconjugation effect is not observed in:

(A) 

(B) 

(C) 

(D) 

5. Phosphorous undergoes disproportionation in alkaline medium as shown below:

$$aP_4(s) + bOH^-(aq) \rightarrow cPH_3(g) + dH_2PO_2^-(aq)$$

The values of a, b, c and d in the balanced redox reaction are respectively:

(A) 1, 2, 1, 3 **(B)** 1, 3, 1, 3 **(C)** 1, 2, 4, 3 **(D)** 1, 3, 4, 3

6. In acidic medium, MnO_4^- oxidizes H_2S to:

(A) SO_4^{2-} **(B)** SO_2 **(C)** SO_3^{2-} **(D)** S

7. In the Lassaigne's test for nitrogen in an organic compound, the Prussian blue colour is obtained due to formation of:

(A) $Na_4[Fe(CN)_6].xH_2O$ **(B)** $Fe_2[Fe(CN)_6].xH_2O$

(C) $Fe_3[Fe(CN)_6].xH_2O$ **(D)** $Fe_4[Fe(CN)_6]_3.xH_2O$

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8. Match List I with List II

List I (Complexion)		List II (Color)	
P.	$[Co(NH_3)_6]^{3+}$	1.	Violet
Q.	$[Ti(H_2O)_6]^{3+}$	2.	Green
R.	$[Ni(H_2O)_6]^{2+}$	3.	Pale blue
S.	$[Ni(H_2O)_4(en)]^{2+}$	4.	Yellowish orange

Choose the correct answer from the options given below:

(A) P – 1, Q – 2, R – 4, S – 5 **(B)** P – 4, Q – 3, R – 2, S – 1

(C) P – 3, Q – 2, R – 4, S – 1 **(D)** P – 4, Q – 1, R – 2, S – 3

9. $Na_2S + H_2SO_4 \rightarrow Na_2SO_4 + X(gas)$

$Pb^{2+} + X \rightarrow Y(ppt)$

Which of the following statements are correct?

(I) X has rotten egg smell

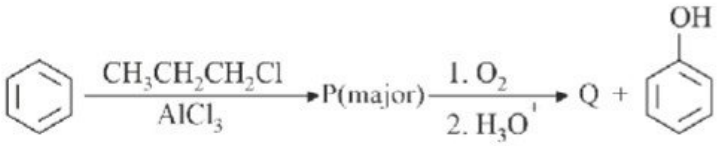
(II) X turns lime water milky

(III) The color of Y is yellow

(IV) The color of Y is black.

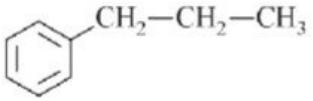
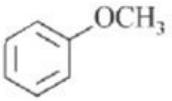
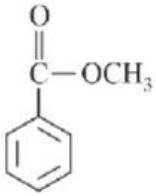
(A) I and III **(B)** I, II and III **(C)** I and IV **(D)** II and III

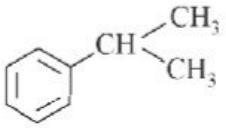
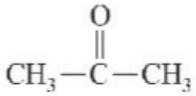
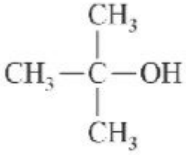
SPACE FOR ROUGH WORK

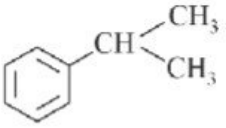
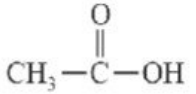
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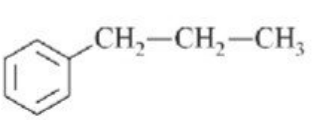
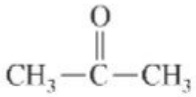
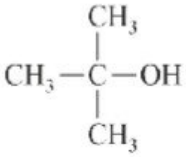
$$Q \xrightarrow[2.H_3O^+]{1.CH_3MgBr} R$$

P, Q and R respectively are:

(A) P.  Q.  R. 

(B) P.  Q.  R. 

(C) P.  Q.  R. CH_4

(D) P.  Q.  R. 

11. The stability of the allyl carbocation is mainly due to:

(A) Inductive effect **(B)** Resonance effect

(C) Hyperconjugation effect **(D)** Electromeric effect.

SPACE FOR ROUGH WORK

12. Given below are two statements:

One is labelled as assertion A and the other is labelled as reason R

Assertion A: The first ionisation enthalpy of Boron (Z = 5) is slightly less than that of Beryllium (Z = 4) even though the former has a greater nuclear charge

Reason R: The penetration of a 2s-electron to the nucleus is less than that of a 2p electron.

In the light of above statements, choose the most appropriate from the options given below:

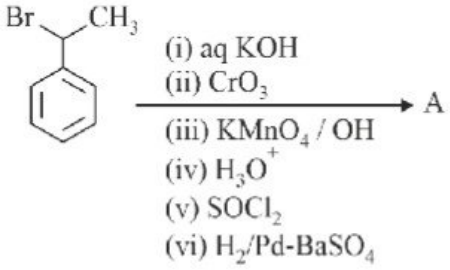
(A) Both A and R are true but R is not the correct explanation of A

(B) A is true but R is false

(C) Both A and R are true and R is the correct explanation of A

(D) A is false but R is true.

13. The product A in the following multistep reaction sequence is:



(A) Benzyl alcohol **(B)** Toluene

(C) Benzaldehyde **(D)** Cyclohexane carbaldehyde

14. **Assertion:** Due to higher electronegativity of sp^2 hybridized carbon of phenol to which –OH is attached, electron density on its oxygen atom is lower than a saturated alcohol.

Reason: In alkoxide ion, negative charge is localised on oxygen while in phenoxide ion, the charge is delocalised.

(A) Both A and R are true but R is not the correct explanation of A

(B) A is true but R is false

(C) Both A and R are true and R is the correct explanation of A

(D) A is false but R is true.

SPACE FOR ROUGH WORK

15. Given below are two statements:

Statement I: First ionisation enthalpy of group 13 elements gradually decreases in magnitude from B to Tl.

Statement II: Galium has unusually low melting point.

In the light of above statements, choose the most appropriate from the options given below:

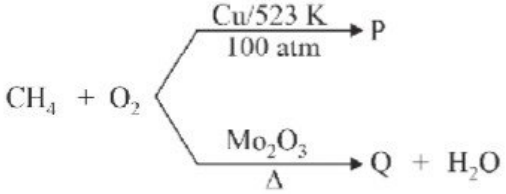
(A) Both Statement I & Statement II are true

(B) Statement I is true but Statement II is false

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are false

16. Identify product P and product Q



(A) P – HCHO Q – HCHO

(B) P – CH_3OH Q – HCH*O*

(C) P – CH_3COOH *Q* – CH_3OH

(D) P – CH_3CO*OH* Q – HCHO

17. C*onsider* the following statements about chromatography.

I. Silica gel can be used as an adsorbent in column chromatography.

II. Thin layer chromatography is a type of adsorption chromatography

III. In paper chromatography, the chromatography paper contains water trapped in it which acts as stationary phase.

The correct statements are:

(A) I and II only **(B)** II and III only **(C)** I and III only **(D)** I, II and III

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18. Osteomalacia is caused by deficiency of which of the following?

(A) Vitamin B_{12} **(B)** Vitamin D **(C)** Vitamin A **(D)** Vitamin K

19. Identify the incorrect match from the following:

(A) $[Fe(NO)]SO_4$ - Brown

(B) $Na_4[Fe(CN)_5NOS]$ - Purple

(C) PbS - Black

(D) $Cr_2(SO_4)_3$ - Red

20. $FeCr_2O_4 + Na_2CO_3 + air \rightarrow A + Fe_2O_3 + CO_2$

$A + H_2SO_4 \rightarrow \underset{(orange)}{B}$

The compound A and B respectively are:

(A) $A \rightarrow Na_2Cr_2O_7$ $B \rightarrow Na_2SO_4$

(B) $A \rightarrow Na_2CrO_4$ $B \rightarrow Na_2Cr_2O_7$

(C) $A \rightarrow NaOH$ $B \rightarrow Na_2SO_4$

(D) $A \rightarrow Na_2CrO_4$ $B \rightarrow Na_2SO_4$

SPACE FOR ROUGH WORK

SECTION-2

Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 (any) questions have to be attempted.

The answer to each question should be **rounded off to the nearest integer**.

1. Consider the following series of reactions:

$$CH_3-CH_2Cl \xrightarrow[\substack{(ii)\,Br_2/CCl_4\\(iii)\,NaNH_2\,(2eq)\\(iv)\,Red\,Hot\,Fe\,Tube\\(v)\,CH_3-\overset{\overset{O}{\|}}{C}-Cl/AlCl_3}]{(i)\,KOH/ethanol} (P)\,Product$$

The total number of sp^2 carbon atoms present per molecule of the product (P) is _____.

2. The number of minutes it will take to plate out 5.2 g of Cr from a $Cr_2(SO_4)_3$ solution using a current of 9.65 A is _____. [Atomic mass of Cr = 52 g/mol]

3. Consider the following reactions:

(I) Decomposition of PCl_5 (g)

(II) Decomposition of gaseous NH_3 on a hot platinum surface

(III) Thermal decomposition of HI on gold surface

(IV) $^{226}_{88}Ra \rightarrow ^{4}_{2}He + ^{222}_{86}Rn$

(V) Decomposition of N_2O_5

(VI) Decomposition of N_2O_4

How many of the above reactions are first order reactions?

4. The number of species from the following which are diamagnetic with bond order equal to one is_____.

$$H_2, He_2^+, O_2^+, N_2^{2-}, O_2^{2-}, F_2, Ne_2^+, B_2$$

5. Two elements X and Y combine to form two compounds XY_2 and X_2Y_3. 0.05 mole of XY_2 weighs 5 g while 3.011×10^{23} molecules of X_2Y_3 weighs 85 g. The sum of atomic masses of X and Y is_____.

SPACE FOR ROUGH WORK

6. How many compounds from the following have higher boiling point than propanol?

(i)	n-Butane	**(ii)**	Methoxyethane	**(iii)**	propanal
(iv)	Isobutane	**(v)**	Butanol	**(vi)**	Ethanal
(vii)	Propanone				

7. When 10 g of a non-volatile substance is dissolved in 105 g of ether the boiling point is raised by 0.5°C. What is the molecular weight of substance? (K_b for the ether is 2.1 K kgmol^{-1})

8. Hydrolysis of sucrose gives:

Sucrose $+ H_2O \rightleftharpoons$ Glucose + Fructose

Equilibrium constant K_C for the reaction is 2×10^{13} at 300K. The value of ΔG° at 300K is $-x \times 10^3$ J/mol (Take R = 8.3 JK^{-1} mol^{-1}). Find the value of x. (Nearest integer)

$(\log_{10} 2 = 0.30 \text{ and } \ln = 2.3\log_{10})$

9. Consider the following species

$SF_4, XeO_3, XeOF_2, ClF_3, I_3^-, XeF_4, BrF_5, H_2O$

Number of species in which hybridization of central atom is sp^3d is _____.

10. X : Number of rings in an anthracene molecule.

Y : Maximum number of π bonds present in an anthracene molecule in its contributor.

Find the value of X + Y.

SPACE FOR ROUGH WORK

PART III: MATHEMATICS	MARKS: 100

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. If $2\sin^2\left((\pi/2)\cos^2 x\right) = 1 - \cos(\pi \sin 2x)$, $x \neq (2n+1)\pi/2, n \in I$, then $\cos 2x$ is equal to:

(A) 1/5 **(B)** 3/5 **(C)** 4/5 **(D)** 1

2. S_1: The absolute minimum value of $|x-1|+|x-2|+|x-3|$ is 2.

S_2: The function $|x-1|+|x-2|+|x-3|$ is differentiable on $R - \{1, 2\}$.

(A) Only S_1 is true **(B)** Only S_2 is true

(C) Both S_1 and S_2 are wrong **(D)** Both S_1 and S_2 are true

3. Let f be a twice differentiable function such that $f''(x) = -f(x)$ and $f'(x) = g(x)$.

If $h'(x) = [f(x)]^2 + [g(x)]^2$, $h(1) = 7$ and $h(0) = 2$, then $h(3)$ is equal to:

(A) 11 **(B)** 4 **(C)** 14 **(D)** 17

4. Two newspapers A and B are published in a city. It is known that 25% of the city population reads A and 20% reads B while 8% reads both A and B. Further, 30% of those who read A but not B look into advertisements and 40% of those who read B but not A also look into advertisements, while 50% of those who read both A and B look into advertisements. Then the percentage of the population who look into advertisements is:

(A) 13.5 **(B)** 13.9 **(C)** 13 **(D)** 128

5. $\lim\limits_{x \to a} \left\{ \left[\left(\dfrac{a^{1/2} + x^{1/2}}{a^{1/4} - x^{1/4}} \right)^{-1} - \dfrac{2(ax)^{1/4}}{x^{3/4} - a^{1/4}x^{1/2} + a^{1/2}x^{1/4} - a^{3/4}} \right]^{-1} - \sqrt{2}^{\log_4 a} \right\}^8$ is:

(A) a **(B)** $a^{3/4}$ **(C)** a^2 **(D)** $a^{1/4}$

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6. Let $P=\begin{bmatrix}1 & 0 & 0\\ 9 & 1 & 0\\ 27 & 9 & 1\end{bmatrix}$ and $Q=\left[q_{ij}\right]_{3\times 3}$ be such that $P^5 - Q = I$, then $\frac{q_{21}+q_{31}}{q_{32}}$ is equal to:

(A) 22 **(B)** 33 **(C)** 44 **(D)** 55

7. The reflection of the point $P(1, 0, 0)$ in the line $\frac{x-1}{2}=\frac{y+1}{-3}=\frac{z+10}{8}$ is:

(A) $(3, -4, -2)$ **(B)** $(5, -8, -4)$ **(C)** $(1, -1, -10)$ **(D)** $(2, -3, 8)$

8. If $A + B$ is a non-singular matrix, then $A - B - A(A + B)^{-1} A + B(A + B)^{-1} B$ equals:

(A) O **(B)** I **(C)** $A - I$ **(D)** $I - B$

9. One hundred identical coins, each with probability p of showing heads are tossed once. If $0 < p < 1$ and the probability of heads showing on 50 coins is equal to that of heads showing on 51 coins, the value of p is:

(A) 1/2 **(B)** 51/101 **(C)** 49/101 **(D)** 3/101

10. A line meets the positive coordinate axes in A and B. A circle is circumscribed about the triangle OAB (where O is origin). If m and n are the distances of the tangent to the circle at the origin from the points A and B respectively, the diameter of the circle is:

(A) $m(m+n)$ **(B)** $m+n$ **(C)** $n(m+n)$ **(D)** $(1/2)(m+n)$.

11. Let $P = (-1, 0)$, $Q = (0, 0)$ and $R=(3, 3\sqrt{3})$ be three points. Then the equation of the bisector of the angle PQR is:

(A) $(\sqrt{3}/2)x+y=0$ **(B)** $x+\sqrt{3}y=0$

(C) $\sqrt{3}x+y=0$ **(D)** $x+(\sqrt{3}/2)y=0$

SPACE FOR ROUGH WORK

12. A vector whose modulus is $\sqrt{51}$ and makes the same angle with $\vec{a}=\frac{\hat{i}-2j+2k}{3}, \vec{b}=\frac{-4\hat{i}-3k}{5}$ and $\vec{c}=j$, will be:

(A) $5\hat{i}+5j+k$ **(B)** $5\hat{i}+j-5k$ **(C)** $5\hat{i}+j+5k$ **(D)** $\pm(5\hat{i}-j-5k)$

13. Let f, g be continuously differentiable functions from R^+ to R^+ then the value of $\int_a^b \frac{f(x)g'(x)-f'(x)}{f(x)+e^{g(x)}}dx$ is equal to:

(A) $\log\left(\frac{e^{-g(b)}f(b)+1}{e^{-g(a)}f(a)+1}\right)$ **(B)** $\log\left(\frac{e^{-g(a)}f(a)+1}{e^{-g(b)}f(b)+1}\right)$

(C) $\log\left(\frac{e^{-g(a)}f(b)+1}{e^{-g(a)}f(a)+1}\right)$ **(D)** $\log\left(\frac{e^{-g(b)}f(a)+1}{e^{-g(a)}f(b)+1}\right)$

14. Let ω be a complex number such that $|\omega|=\sqrt{2}$. Let α, β be solution of the equation $z^2+i(\omega+i\bar{\omega})z-2i=0$, then:

(A) $|\alpha|+|\beta|=2$ **(B)** $|\alpha-\beta|=2$ **(C)** $|\alpha|+|\beta|=2\sqrt{2}$ **(D)** $|\alpha-\beta|=2\sqrt{2}$

15. If $I=\int\frac{dx}{\sqrt{(5+2x+x^2)^3}}$,

$=A\left(\frac{x+1}{\sqrt{5+2x+x^2}}\right)+C$ then A is equal to:

(A) $\frac{1}{2}$ **(B)** 2 **(C)** $\frac{1}{4}$ **(D)** $\frac{4}{3}$

16. The domain of definition of

$f(x)=\sqrt{\log_{0.4}\left(\frac{x-1}{x+5}\right)}\times\frac{1}{x^2-36}$ is:

(A) $(-\infty,0)\sim\{-6\}$ **(B)** $(0,\infty)\sim\{1,6\}$ **(C)** $(1,\infty)\sim\{6\}$ **(D)** $[1,\infty)\sim\{6\}$

SPACE FOR ROUGH WORK

17. If $f(1) = 1$, $f'(1) = 3$ then the derivative of $y = f(f(f(f(x))))$ at $x = 1$ is:

(A) 256 **(B)** 16 **(C)** 81 **(D)** 27

18. S_1: The locus of a point which is equidistant from the points whose position vectors are $3\hat{i} - 2j + 5k$ and $\hat{i} + 2j - k$ is $\vec{r} \cdot (\hat{i} - 2j + 3k) = 8$.

S_2: The locus of a point which is equidistant from the points whose position vectors are $\boldsymbol{a}$ and $\mathbf{b}$ is $\left(\vec{r} - \frac{\vec{a} + \vec{b}}{2}\right).(\vec{a} - \vec{b}) = 0$

(A) S_1 and S_2 both are true **(B)** Only S_1 true

(C) Only S_2 true **(D)** S_1 and S_2 both are wrong

19. If $b_1, b_2, b_3 (b_1 > 0)$ are three successive terms of G.P. with common ratio r, the value of r for which the inequality $b_3 > 4b_2 - 3b_1$, holds is:

(A) 1.5 **(B)** 2

(C) 2.5 **(D)** 3.5

20. a, b, c are in A.P., and a^2, b^2, c^2 are in H.P., then which of the following is not correct?

(A) $a = b = c$ **(B)** $a, b, -\frac{1}{2}c$ are in G.P.

(C) a, b, $2c$ are in H.P. **(D)** $-\frac{1}{2}a, b, c$ are in G.P.

SPACE FOR ROUGH WORK

SECTION-2

Section 2 contains 10 Numerical Value Type Questions Out of which ONLY 5 (any) questions have to be attempted.
The answer to each question should be **rounded off to the nearest integer**.

1. If α and β are the roots of the equation $375x^2 - 25x - 2 = 0$, then

$\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\alpha^r + \lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\beta^r$ is equal to $\frac{m}{n}$ where m and n are relatively prime then the value of $m+n$ is:

2. For the hyperbola $H: x^2 - y^2 = 1$ and the ellipse $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b > 0$, let the

(i) Eccentricity of E be reciprocal of the eccentricity of H, and

(ii) The line $y = \sqrt{\frac{5}{2}}x + K$ be a common tangent of E and H Then $4(a^2 + b^2)$ is equal to _____.

3. If p and q be the longest and the shortest distances respectively of the point $(-7, 2)$ from any point (α, β) on the circle $x^2 + y^2 - 10x - 14y - 51 = 0$. If s is the geometric mean of p and q, then $\frac{s}{\sqrt{11}}$ is equal to_____.

4. The value of $y\left(\sqrt{8}\right) - \frac{1}{9}$ if $(1+x^2)\frac{dy}{dx} = x(1-y), y(0) = 4/3$ is_____.

5. Let $X_1, X_2, \ldots, X_{18}$ be eighteen observations such that $\sum\limits_{i=1}^{18}(X_i - \alpha) = 36$ and $\sum\limits_{i=1}^{18}(X_i - \beta)^2 = 90$, where α and β are distinct real numbers. If the standard deviation of these observations is 1, then the value of $|\alpha - \beta|$ is_____.

SPACE FOR ROUGH WORK

6. Let λ be an integer. If the shortest distance between the lines $x-\lambda=2y-1=-2z$ and $x=y+2\lambda=z-\lambda$ is $\frac{\sqrt{7}}{2\sqrt{2}}$, then the value of $|\lambda|$ is _____.

7. Let $(2x^2+3x+4)^{10}=\sum_{r=0}^{20} a_r x^r$. Then $\frac{a_7}{a_{13}}$ is equal to _____.

8. If m and n respectively are the number of local maximum and local minimum points of the function $f(x)=\int_0^{x^2}\frac{t^2-5t+4}{2+e^t}dt$, then $m+n$ is_____.

9. A natural number has prime factorization given by $n=2^x3^y5^z$, where y and z are such that $y+z=5$ and $y^{-1}+z^{-1}=\frac{5}{6}, y>z$. Then the number of odd divisors of n, including 1, is_____.

10. The sum of the squares of the lengths of the chords intercepted on the circle, $x^2+y^2=16$, by the lines, $x+y=n$. $n\in N$, where N is the set of all natural numbers, is_____.

SPACE FOR ROUGH WORK

End of 2024 | Mock JEE Main - 7 (CBT)

Solutions to Mock JEE MAIN – 7 (CBT) | JEE 2024

PHYSICS

SECTION-1

1.(A) As, $\frac{v_1}{v_2}=\frac{8}{27}$; $\frac{r_1}{r_2}=?$

Using law of conservation of linear momentum, $0=m_1v_1-m_2v_2$

(As both are moving in opposite directions.)

Or $\frac{m_1}{m_2}=\frac{v_2}{v_1}=\frac{27}{8}$ or $\frac{\rho\left(\frac{4}{3}\pi r_1^3\right)}{\rho\left(\frac{4}{3}\pi r_2^3\right)}=\frac{27}{8}$ $\therefore$ $\frac{r_1}{r_2}=\frac{3}{2}$

2.(C) de-Broglie wavelength

$\lambda=\frac{h}{p}$ $\therefore \frac{\lambda_1}{\lambda_2}=\frac{p_2}{p_1}$

Since momentum p is conserved in the decay process, $p_2=p_1$

$\therefore$ $\frac{\lambda_1}{\lambda_2}=1$

3.(C) A-S, B-Q, C-R, D-R

(A) $F=6\pi\eta rv$

$\eta=\frac{F}{6\pi rv}=\frac{[MLT^{-2}]}{[L][LT^{-1}]}=[ML^{-1}T^{-1}]$

(B) Surface Tension, $\sigma=\frac{F}{L}=\frac{[MLT^{-2}]}{[L]}$

$=[ML^0T^{-2}]$

(C) Modulus of Elasticity $=\frac{\text{stress}}{\text{strain}}=\frac{\frac{F}{A}}{\frac{\Delta L}{L}}$

$=\frac{\frac{[MLT^{-2}]}{[L^2]}}{\frac{[L]}{[L]}}=[ML^{-1}T^{-2}]$

(D) Energy per unit volume ≡ Pressure

$\equiv\frac{[MLT^{-2}]}{[L^2]}=[ML^{-1}T^{-2}]$

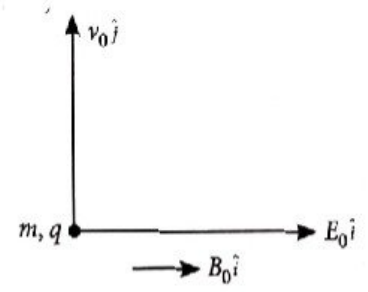
4.(A) Mass of the particle = m, charge = q and initial velocity, $\vec{v}=v_0\hat{j}$

Hence, final velocity will be $v_f^2=v_x^2+v_y^2$

$\Rightarrow$ $(2v_0)^2=v_x^2+v_0^2$

$\Rightarrow$ $v_x^2=4v_0^2-v_0^2=3v_0^2$

$\Rightarrow$ $v_x=\sqrt{3}v_0$



Along x-direction, $F = qE_0 \Rightarrow ma = qE_0$ (Speed does not change in $y-z$ plane)

$$\therefore \qquad a = \frac{qE_0}{m}$$

Now, from equation of motion,

$$v = u + at$$

$$v_x = 0 + \frac{qE_0}{m}t, \sqrt{3}v_0 = 0 + \frac{qE_0}{m}t \qquad \Rightarrow \qquad t = \frac{\sqrt{3}mv_0}{qE_0}$$

5.(B) For convex lens, power

$$P_1 = \frac{100}{40} = 2.5D$$

For concave lens,

$$\text{power } P_2 = -\frac{100}{25} = -4.0D$$

$\therefore$ Power of combination $= P_1 + P_2 = 2.5 - 4.0 = -1.5D$

6.(B) $$F_x = -\left.\frac{\partial U}{\partial X}\right|_{\left(0,\frac{\pi}{4}\right)} = \sin(x+y) = \frac{1}{\sqrt{2}}$$

$$F_y = -\left.\frac{\partial U}{\partial Y}\right|_{\left(0,\frac{\pi}{4}\right)} = \sin(x+y) = \frac{1}{\sqrt{2}}$$

$$\vec{F} = \frac{1}{\sqrt{2}}(\hat{i} + j)$$

7.(B) $$m = \frac{f}{f-u}$$

$$3 = \frac{-24}{-24-u}$$

Or $\quad -24 - u = -8$ or $u + 24 = 8$

Or $\quad u = (8-24)cm = -16cm$

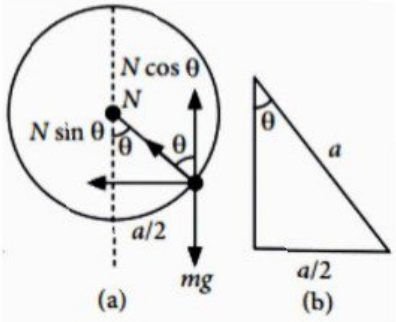
If $m = -3$, then $-3 = \dfrac{-24}{-24-u}$

$$u + 24 = -8$$

Or $\quad u = -32cm$

Note that the magnification is greater than 1, so mirror cannot be convex.

8.(C)



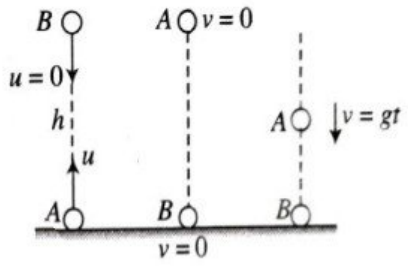
$$N\cos\theta = mg \qquad \ldots(i)$$

$$N\sin\theta = \frac{m\omega^2 a}{2} \qquad \ldots(ii)$$

From Eqs. (ii)/(i), $\tan\theta = \dfrac{\omega^2 a}{2g} \Rightarrow \omega^2 = \dfrac{2g\tan\theta}{a}$

Now, $\sin\theta = \dfrac{a}{2}\times\dfrac{1}{a} = \dfrac{1}{2}$ or $\theta = 30^\circ$ $\quad\therefore\quad$ $\omega^2 = \dfrac{2g}{\sqrt{3}a}$

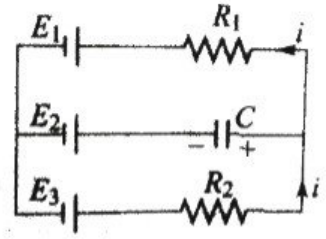
9.(C)



$\vec{V}_{A,B} = \vec{V}_A - \vec{V}_B = (u - gt) - (0 - gt) = u$ = constant

It will remain constant till B reaches ground. B stops on colliding with ground, and A comes down with increasing velocity.

10.(C)



$i = \dfrac{2.5-1}{30} = 0.05A$

$-20i + 2.5 = V_{\text{capacitor}} + 1$

$V_{\text{capacitor}} = 0.5V$

$q^{(-)} = CV = 10\times 0.5 = 5\mu C$

11.(A) $\rho = \dfrac{m}{\pi r^2 l}$

$\Rightarrow \quad \left(\dfrac{\Delta\rho}{\rho}\times 100\right) = \left(\dfrac{\Delta m}{m} + \dfrac{2\Delta r}{r} + \dfrac{\Delta l}{l}\right)\times 100 = \left(\dfrac{0.003}{0.3} + \dfrac{2\times 0.005}{0.5} + \dfrac{0.06}{6}\right)\times 100 = 4\%$

12.(D) From Dalton's law, final pressure of the mixture of nitrogen and oxygen is

$P = P_1 + P_2$

$= \dfrac{n_1 RT}{V} + \dfrac{n_2 RT}{V} = \dfrac{m_1}{M_1}\dfrac{RT}{V} + \dfrac{m_2}{M_2}\dfrac{RT}{V} = \dfrac{8}{32}\dfrac{RT}{V} + \dfrac{7}{28}\dfrac{RT}{V} = \dfrac{RT}{2V}$

$\Rightarrow \quad 10 = \dfrac{RT}{2V}$ …(i)

When oxygen is absorbed then for nitrogen let pressure is

$P = \dfrac{7}{28}\dfrac{RT}{V} \quad \Rightarrow \quad P = \dfrac{RT}{4V}$ …(ii)

From Eqs. (i) and (ii), we get pressure of the nitrogen P = 5 atm

13.(B) From work-energy theorem, net work done by all forces (internal and external) = change in kinetic energy. $\Rightarrow$ Work done by gravity + work done by air drag

$= \dfrac{1}{2}\times 70(20^2 - 0^2)$ $\Rightarrow$ Work done by air drag = 14000 – 35000 = –21000 J

14.(B) $\vec{E} = 8\hat{i} + 4j + 3k, \qquad \vec{A} = 100k$

So, $\phi = \vec{E} \cdot \vec{A} = 300$ units

15.(D) Here, angular momentum is conserved. According to it,

$I_1\omega_1 = I_2\omega_2$

Or $MR_1^2\omega_1 = MR_2^2\omega_2$ or $R_1v_1 = R_2v_2$

At point P_4, the value of R is minimum and hence the velocity is maximum or KE is maximum.

16.(C) $U = -PE\cos\theta$

As θ increases U increases

17.(A) Above 6 V the zener diode breakdown. So, the current through diode

$I_Z = I_i - I_L$

Or $I_Z = \dfrac{V_B - V_Z}{R_i} - \dfrac{V_Z}{R_L}$

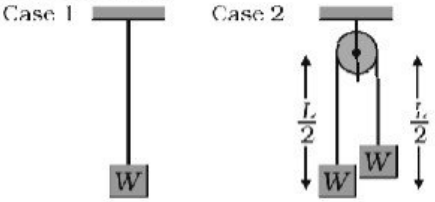
For minimum current,

$I_{min} = \dfrac{8-6}{1} - \dfrac{6}{4} = 0.5mA$

For maximum current,

$I_{max} = \dfrac{16-6}{1} - \dfrac{6}{4} = 8.5mA$

18.(A) Let us consider the length of wire as L and cross-sectional area A, the material of wire has Young's modulus as Y.



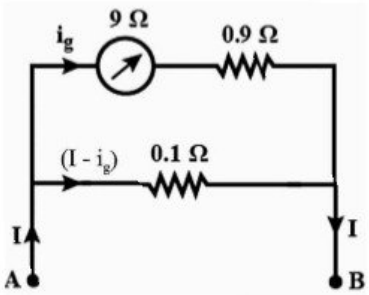
Then, for 1st case, $Y = \dfrac{w/A}{l/L}$ For 2nd case, $Y = \dfrac{w/A}{2l'/L} \Rightarrow l' = \dfrac{l}{2}$

So, total elongation of both sides $= 2l' = l$

19.(C) $i_g = 10mA = 0.01A$

$V_A - V_B = (I - i_g)0.1 = i_g(9 + 0.9)$

Or $I = \dfrac{10\times10^{-3}\times10}{0.1} = 1A$



20.(B) Here $R = X_L = X_C$ ($\because$ Voltage across them is same)

When capacitor is short circuited,

$$I = \frac{10}{\left(R^2 + X_L^2\right)^{1/2}} = \frac{10}{\sqrt{2}R}$$

$\therefore$ Potential drop across inductance = $IX_L = IR = 10/\sqrt{2}V$

SECTION-2

1.(106)

$$v = \sqrt{\frac{\gamma P}{\rho}}$$

$$\frac{v_{gas}}{v_{air}} = \sqrt{\frac{\rho_{air}}{\rho_{gas}}} = \sqrt{\frac{1}{4}} \qquad \Rightarrow v_{gas} = \frac{300}{2} = 150 \text{ m/s}$$

So, $n_2 - n_1 = \frac{v_{gas}}{2l} = \frac{150}{2\times 1} = 75$

Or $\Delta n = 75Hz$

2.(15) $V = 300\sqrt{2}\sin(50t)$, $V_0 = 300\sqrt{2}$,

$\omega = 50 \text{ rad/s}$

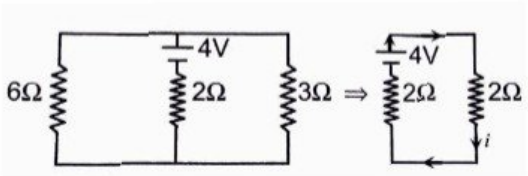
$I_{rms} = V_{rms} / X_c = V_{rms} \times \omega C$

$$= \frac{V_0}{\sqrt{2}} \times \omega C = \frac{300\sqrt{2}}{\sqrt{2}} \times 50 \times 1 \times 10^{-6} = 15000 \times 10^{-6} = 15 \text{mA}$$

3.(2) Motional emf $E = Bvl$

$E = (2)(2)(1) = 4\text{V}$

This acts as a cell of emf E = 4 V and internal resistance $r = 2\Omega$. The simple circuit can be drawn as follows:



$\therefore$ Current through the connector

$$i = \frac{4}{2+2} = 1A$$

Magnetic force on connector $F_m = ilB$ = (1) (1) (2) = 2 N (towards left)

Therefore, to keep the connector moving with a constant velocity, a force of 2 N will have to be applied towards right.

4.(5) Given trajectory equation: $y = x - \frac{x^2}{20}$

Comparing with standard projectile equation

$$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta};$$

θ : angle of projection

u : initial speed of projection

$$\tan\theta = 1 \Rightarrow \theta = 45^\circ$$

$$\frac{2u^2\cos^2\theta}{g}=20$$

$$\Rightarrow \quad u^2=\frac{20\times g}{2\times\cos^2 45^\circ}=20g \qquad \therefore \qquad H_{max}=\frac{u^2\sin^2\theta}{2g}=\frac{20\times g\times\sin^2 45^\circ}{2g}$$

5.(56) $\vec{S}_{c.m}=\vec{u}_{c.m}t+\frac{1}{2}\vec{a}_{c.m}\times t^2$

Put $S_{c.m}=0$

$$a_{c.m}=\frac{g\sin\theta}{\left(1+\frac{I_{c.m}}{mR^2}\right)}=\frac{10\times\sin 30^\circ}{1+\left(\frac{\frac{2}{5}mR^2}{mR^2}\right)}=\frac{25}{7}m/s^2$$

$$O=1\times t-\frac{1}{2}\times\frac{25}{7}\times t^2 \quad \Rightarrow 25t^2-14t=0\Rightarrow t=\frac{14}{25}\simeq 0.56\sec$$

6.(40) Heat lost by the steam = Heat gained by the ice

$$M\times 540+M\times 1\times(100-40)=200\times 80+200(40-0); \qquad M=40\,gm$$

7.(32) $\vec{F}_{AOB}=\vec{F}_{AB}=i(\vec{l}\times\vec{B})$

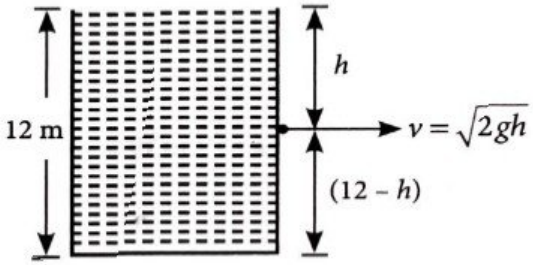
Here, $AB=2\sqrt{2\times 2}=4m \qquad \therefore \qquad \vec{F}_{AB}=2\left[(-4j)\times(-4k)\right]=32\hat{i}$

8.(18) Fringe width, $\omega=\frac{\lambda D}{d}\propto\lambda$

When the wavelength is decreased from 600 nm to 400 nm, fringe width will also decrease by a factor of $\frac{4}{6}$ or $\frac{2}{3}$ or the number of fringes in the same segment will increase by a factor of 3/2.

Therefore, number of fringes observed in the same segment $=12\times\frac{3}{2}=18$

9.(6) The velocity of efflux is, $v=\sqrt{2gh}$



After emerging from the orifice, the water adopts parabolic path. Let it takes time t sec in covering a horizontal range R.

Then, $12-h=\frac{1}{2}gt^2\Rightarrow t=\sqrt{\frac{2(12-h)}{g}}$

Range, $R=vt=\sqrt{2gh}\sqrt{\frac{2(12-h)}{g}} \qquad \Rightarrow \qquad R=\sqrt{4h(12-h)}$

For maximum range $\frac{dR}{dh}=0 \qquad \Rightarrow \qquad h=6m$

10.(136) Given, length of the wire, l = 1 m

Current in the wire, I = 1 A

Cross-section area of the wire,

A = 2.0 mm^2 = 2.0 × 10^{-6} m^2

Resistivity, ρ = 1.7 × 10^{-8} Ω m

We know that $R = \rho \frac{l}{A}$

$$\therefore \quad R = \frac{1.7\times10^{-8}\times1}{2.0\times10^{-6}} = 0.85\times10^{-2}\Omega$$

Potential difference across the wire, V = IR

=1 × 0.85 × 10^{-2} = 0.85 × 10^{-2} V

Force experienced by a moving electron in the wire will be,

$$F = qE = \frac{qV}{l} = \frac{1.6\times10^{-19}\times0.85\times10^{-2}}{1} = 1.36 \times 10^{-21} = 136 \times 10^{-23}\ N$$

CHEMISTRY

SECTION-1

1.(B) Higher the value of 'n + l', greater is the energy of the electron.
For
(A) $n + l = 3 + 2 = 5$
(B) $n + l = 4 + 3 = 7$
(C) $n + l = 4 + 1 = 5$
(D) $n + l = 5 + 0 = 5$

2.(C) Statement (C) is false.
The M-C π bond is formed by the donation of a pair of electrons from filled d orbitals of metal into vacant π^* orbital of CO.

3.(C) $\Delta H = \Delta U + \Delta n_g RT$

For $\Delta n_g > 0$, $\Delta H > \Delta U$

For, $FeCO_3 \rightarrow FeO(s) + CO_2(g)$

$\Delta n_g = 1$

$\therefore$ $\Delta H > \Delta U$

4.(A) Hyperconjugation doesn't take place in (A)

5.(B) $P_4 + 3OH^- + 3H_2O \rightarrow PH_3 + 3H_2PO_2^-$

6.(D) $H_2S \rightarrow 2H^+ + S^{2-}$

$5S^{2-} + 2MnO_4^- + 16H^+ \rightarrow 2Mn^{2+} + 8H_2O + 5S.$

7.(D) $6CN^- + Fe^{2+} \rightarrow [Fe(CN)_6]^{4-}$

$3[Fe(CN)_6]^{4-} + 4Fe^{3+} \xrightarrow{xH_2O} Fe_4[Fe(CN)_6]_3.xH_2O$
Prussian blue

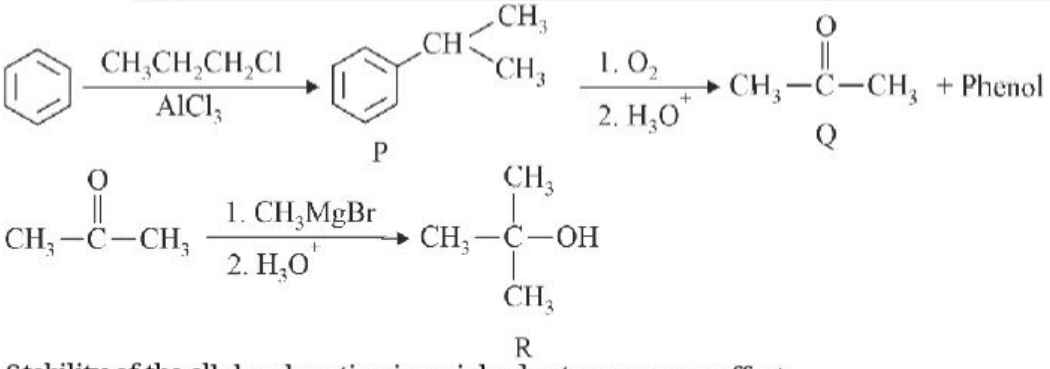
8.(D)

List I (Complex ion)		List II (Color)	
P.	$[Co(NH_3)_6]^{3+}$	4	Yellowish orange
Q.	$[Ti(H_2O)_6]^{3+}$	1.	Violet
R.	$[Ni(H_2O)_6]^{2+}$	2.	Green
S.	$[Ni(H_2O)_4(en)]^{2+}$	3.	Pale blue

P – 4, Q – 1, R – 2, S – 3

9.(C) $X \rightarrow H_2S$ (rotten egg smell)
$Y \rightarrow PbS$ (black ppt)

10.(B)



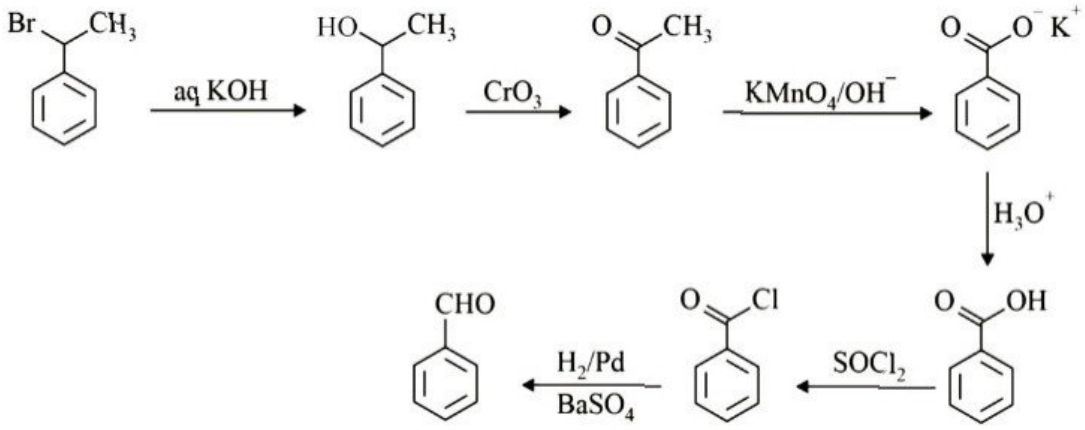
11.(B) Stability of the allyl carbocation is mainly due to resonance effect.

12.(B) A is true

R is false

The penetration of a 2s-electron to nucleus is more than that of a 2p-electron.

13.(C)



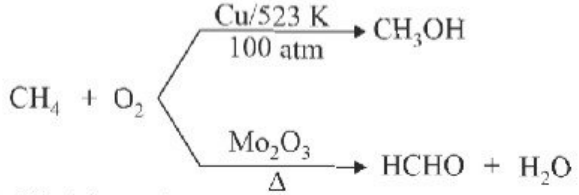
Benzaldehyde

14.(A) Both A and R are true but R is not the correct explanation of A

15.(C) Order of Ionisation enthalpy

: B > Al < Ga > In < Tl

16.(B)



17.(D) All statements are correct.

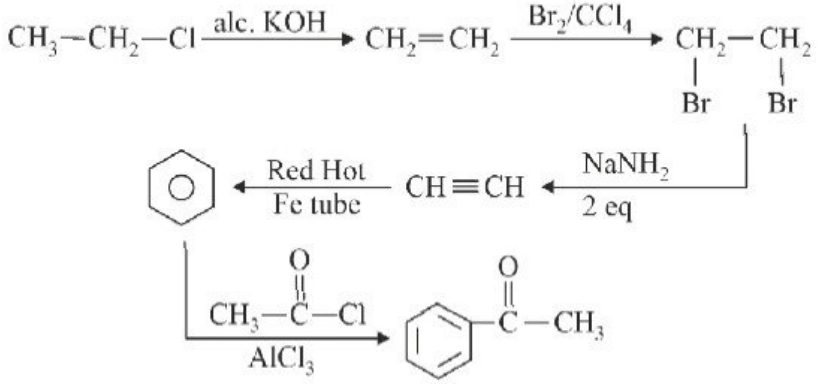
18.(B) Osteomalacia is caused by deficiency of Vitamin D.

19.(D) $Cr_2(SO_4)_3 \rightarrow$ green

20.(B) $A \rightarrow Na_2CrO_4$ $B \rightarrow Na_2Cr_2O_7$

SECTION-2

1.(7)



2.(50) W = Z it

$$5.2 = \frac{52}{3} \times \frac{9.65}{96500} \times t$$

$$\Rightarrow t = 3000\,\text{sec}$$

Or 50 min

3.(4) I, IV, V, VI – First order reactions
II, III – Zero order reactions

4.(3) H_2, O_2^{2-}, F_2 are diamagnetic with bond order = 1

5.(70) Mol mass of $XY_2 = \frac{5}{0.05} = 100$

Mol mass of $X_2Y_3 = \frac{85}{3.011 \times 10^{23}} \times 6.022 \times 10^{23}$

= 170

Let molar mass of X = a

Molar mass of Y = b

$\therefore \quad a + 2b = 100, 2a + 3b = 170 \quad \Rightarrow \quad a = 40, b = 30$

6.(1) (v) has higher b.p. than propanol.

7.(400) $\Delta T_b = mK_b$

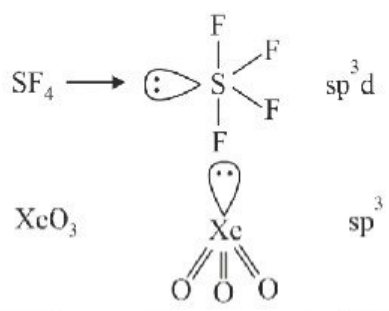
$0.5 = \left(\frac{10}{M}\right)\frac{1000}{105} \times 2.1$ $\quad \therefore$ Molal boiling point constant for ether is 2.1°C kg/mol.

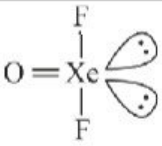
$$M = \frac{10 \times 1000}{0.5} \times \frac{2.1}{105} = 400$$

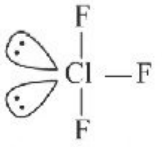
8.(76) $\Delta G^\circ = -RT \ln K_C = -8.3 \times 300 \times \ln(2 \times 10^{13})$

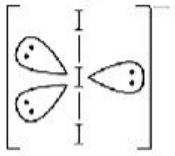
$= -8.3 \times 300 \times 2.3 \times \log(2 \times 10^{13}) = -8.3 \times 300 \times 2.3 \times [0.3 + 13] = -76 \times 10^3\,\text{J mol}^{-1}$

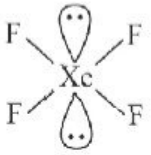
9.(4)

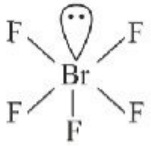


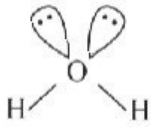
$XeOF_2$  sp^3d

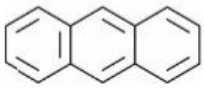
ClF_3  sp^3d

I_3^-  sp^3d

XeF_4  sp^3d^2

BrF_5  sp^3d^2

H_2O  sp^3

10.(10) Anthracene is 

X = 3
Y = 7
X+Y = 10

MATHEMATICS

SECTION-1

1.(B) The given equation is equivalent to

$$2\sin^2\left((\pi/2)\cos^2 x\right) = 2\sin^2((\pi/2)\sin 2x)$$

$$\Rightarrow \quad \cos^2 x = \sin 2x \quad \Rightarrow \quad \cos x(\cos x - 2\sin x) = 0$$

$$\Rightarrow \quad 1 - 2\tan x = 0 \text{ as } \cos x \neq 0,$$

$$\because \quad x \neq (2n+1)\pi/2$$

$$\Rightarrow \quad \tan x = 1/2 \quad \Rightarrow \quad \cos 2x = \frac{1-\tan^2 x}{1+\tan^2 x} = \frac{3}{5}.$$

2.(A)

$$f(x) = |x-1| + |x-2| + |x-3| = \begin{cases} 6-3x & , x \le 1 \\ 4-x & , 1 < x \le 2 \\ x & , 2 < x \le 3 \\ 3x-6 & , x > 3 \end{cases}$$

The function has absolute minimum at x = 2. So, the absolute minimum is equal

$$2.\ f'(1^-) = -3,\ f'(1^+) = -1$$

$$f'(2^+) = 1,\ f'(2^-) = -1,\ f'(3^-) = 1,\ f'(3^+) = 3.$$

Hence, f is not differentiable at x = 1, 2, 3.

3.(D) $h'(x) = \left[f(x)\right]^2 + \left[g(x)\right]^2$

$$h''(x) = 2f(x)\,f'(x) + 2g(x)\,g'(x)$$

$$= 2f(x)\,g(x) + 2g(x)\,f''(x) \qquad [\because f'(x) = g(x)]$$

$$= 2f(x)g(x) - 2g(x)\,f(x) = 0 \left[\because f''(x) = -f(x)\right]$$

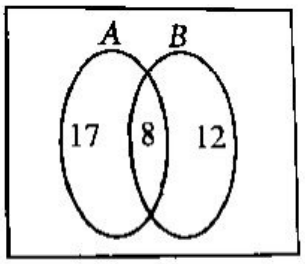
Thus h'(x) = k, a constant, for all $x \in R$. Hence h(x) = kx + m, so that from h(0) = 2, we get $m = 2$ and from h(1) = 7, we get k = 5. Therefore, h(3) = 17.

4.(B) Let the total population is 100.

$$n\left(A \cap \overline{B}\right) = 17 \text{ and } n\left(\overline{A} \cap B\right) = 12$$

$$\therefore \quad n(A) = 25,$$

$$n(B) = 20, n(A \cap B) = 8,$$



Now 40% of $n(\overline{A} \cap B) = 4.8$ look advertisement.

30% of $n(A \cap \overline{B}) = 5.1$ look into advertisement.

50% of $n(A \cap B) = 4$ look into advertisement.

$\therefore$ Total percentage of population, who look into advertisement = 4.8 + 5.1 + 4 = 13.9.

5.(C) Simplifying the expression in brackets by setting $a^{1/4}$ = b and $x^{1/4}$ = y, the function whose limit is required can be written as

$$\left\{\left[\left(\frac{b^2+y^2}{b-y}\right)^{-1} - \frac{2by}{y^3 - by^2 + b^2y - b^3}\right]^{-1} - 2^{\frac{1}{2}\log_4 b^4}\right\}^8$$

$$=\left\{\left[\frac{b-y}{b^2+y^2}-\frac{2by}{(y-b)(y^2+b^2)}\right]^{-1}-b\right\}^8=\left\{\left[\frac{1}{b-y}\right]^{-1}-b\right\}^8=y^8=x^2.$$

Hence the required limit as $x\to a$ is a^2.

6.(A) Write P = I + R where

$$R=\begin{bmatrix}0&0&0\\9&0&0\\27&9&0\end{bmatrix}$$

$$R^2=\begin{bmatrix}0&0&0\\0&0&0\\81&0&0\end{bmatrix},R^3=0$$

$\Rightarrow R^k=0\ \forall\ k\geq 3.$

We have $P^5=(I+R)^5=I+5R+10R^2$

$$\Rightarrow Q=P^5-I=5R+10R^2=\begin{bmatrix}0&0&0\\45&0&0\\945&45&0\end{bmatrix}$$

$$\frac{q_{21}+q_{31}}{q_{32}}=\frac{45+945}{45}=22$$

7.(B) Coordinates of any point Q on the given line are

$(2r + 1, -3r - 1, 8r - 10)$.

So, the direction ratios of PQ are $2r, -3r-1, 8r-10$.

Now PQ is perpendicular to the given line.

if $2(2r) - 3(-3r - 1) + 8(8r - 10) = 0$

$\Rightarrow$ $77r - 77 = 0 \Rightarrow r = 1$

and the coordinates of Q, the foot of the perpendicular from P on the line are $(3, -4, -2)$.

Let $R(a, b, c)$ be the reflection of P in the given lines then Q is the mid-point of PR

$$\Rightarrow \quad \frac{a+1}{2}=3,\frac{b}{2}=-4,\frac{c}{2}=-2$$

$\Rightarrow$ $a = 5, b = -8, c = -4$

and the coordinate of the required point are $(5, -8, -4)$.

8.(A) $A - B - A(A + B)^{-1} A + B(A + B)^{-1} B$

$= A[I - (A + B)^{-1} A] - [I - B(A + B)^{-1}]B$

$= A(A + B)^{-1} [A + B - A] - [A + B - B] (A + B)^{-1} B$

$= A(A + B)^{-1} B - A (A + B)^{-1} B = O$

9.(B) Let X be the number of coins showing heads.

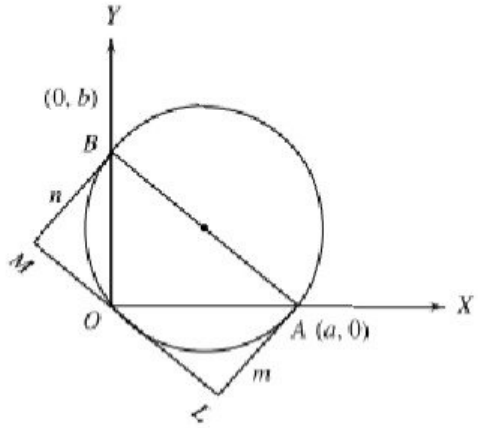
Then X follows a binomial distribution with parameters $n = 100$ and p. Since $P(X = 50) = P(X = 51)$, we get

$$^{100}C_{50}p^{50}(1-p)^{50}={}^{100}C_{51}p^{51}(1-p)^{49}$$

$$\Rightarrow \quad \frac{100!}{50!50!}\cdot\frac{51!49!}{100!}=\frac{p}{1-p}\Rightarrow\frac{51}{50}=\frac{p}{1-p}$$

$$\Rightarrow \quad 51-51p=50p\Rightarrow p=51/101$$

10.(B)



Let the coordinates of A be (a, 0) and that of B be (0, b). Since $\angle AOB = \pi/2$, the line AB is a diameter of the circle circumscribing the triangle OAB, its centre is the mid-point of AB, i.e., (a/2, b/2), and its radius is (1/2) AB = $(1/2)\sqrt{a^2+b^2}$. Therefore, equation of the circle through O, A and B is $x^2 + y^2 - ax - by = 0$, and the equation of the tangent at the origin to this circle is $ax + by = 0$. If AL and BM are the perpendiculars from A and B to this tangent, then

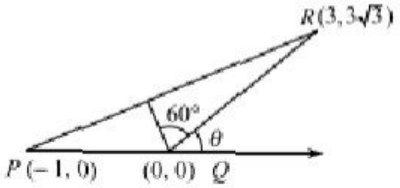
$$Al = m = \frac{a^2}{\sqrt{a^2+b^2}} \text{ and } BM = n = \frac{b^2}{\sqrt{a^2+b^2}}$$

Hence $m + n = \sqrt{a^2+b^2}$ is the diameter of the circle.

11.(C) Slope of $RQ = \sqrt{3} = \tan\theta$

$\Rightarrow \theta = 60^\circ \Rightarrow PQR = 120^\circ$

so bisector of angle PQR makes 120° with the +ve direction of x-axis and passes through Q(0,0) and hence its equation is $y = -\sqrt{3}x$ or $\sqrt{3}x + y = 0$



12.(D) Let the required vector be $\vec{\alpha} = d_1\hat{i} + d_2 j + d_3 k$,

where $d_1^2 + d_2^2 + d_3^2 = 51$, (given) ...(i)

Now, each of the given vectors a, b, c is a unit vector

$$\cos\theta = \frac{\vec{d}.\vec{a}}{|\vec{d}||\vec{a}|} = \frac{\vec{d}.\vec{b}}{|\vec{d}||\vec{b}|} = \frac{\vec{d}.\vec{c}}{|\vec{d}||\vec{c}|}$$

Or $\vec{d}.\vec{a} = \vec{d}.\vec{b} = \vec{d}.\vec{c}$

$|\vec{d}| = \sqrt{51}$ cancels out and $|\vec{a}| = |\vec{b}| = |\vec{c}| = 1$

Hence, $\frac{1}{3}(d_1 - 2d_2 + 2d_3) = \frac{1}{5}(-4d_1 + 0d_2 - 3d_3) = d_2$

$\Rightarrow d_1 - 5d_2 + 2d_3 = 0$ and $4d_1 + 5d_2 + 3d_3 = 0$

On solving, $\frac{d_1}{5}=\frac{d_2}{-1}=\frac{d_3}{-5}=\lambda$ (say)

Putting d_1, d_2 and d_3 in (i), we get $\lambda=\pm1$

Hence the required vectors are $\pm(5\hat{i}-j-5k)$.

13.(B) The given integral can be written as

$$\int_a^b \frac{e^{-g(x)}f(x)g'(x)-f'(x)e^{-g(x)}}{f(x)e^{-g(x)}+1}dx \text{ Let } t=f(x)e^{-g(x)}$$

$$=-\int_{e^{-g(a)}f(a)}^{e^{-g(b)}f(b)}\frac{dt}{t+1}=-\log(t+1)\Big|_{e^{-g(a)}f(a)}^{e^{-g(b)}f(b)}=-\log\left(\frac{e^{-g(b)}f(b)+1}{e^{-g(a)}f(a)+1}\right)=\log\left(\frac{e^{-g(a)}f(a)+1}{e^{-g(b)}f(b)+1}\right)$$

14.(C) $z^2+i(\omega+i\bar{\omega})z-i\omega\bar{\omega}=0 \qquad [\because \omega\bar{\omega}=2]$

$\Rightarrow \quad z(z+i\omega)-\bar{\omega}(z+i\omega)=0 \quad \Rightarrow \quad (z-\bar{\omega})(z+i\omega)=0$

$\Rightarrow \quad z=\bar{\omega},-i\omega$

Let $\alpha=\bar{\omega},\beta=-i\omega$, then

$$|\alpha|+|\beta|=|\bar{\omega}|+|-i\omega|=|\omega|+|\omega|=2\sqrt{2}$$

15.(C) $5+2x+x^2=4+(x+1)^2$. Put $x+1=t$

$I=\int\frac{dt}{\left(4+t^2\right)^3}$. Put $t=2\tan z$, so that

$\sqrt{4+t^2}=2\sec z$ and $dt=2\sec^2 z\,dz$,

$$I=\int\frac{2\sec^2 z\,dz}{2^3\sec^3 z}=\frac{1}{4}\int\cos z\,dz=\frac{1}{4}\sin z+c=\frac{1}{4}\frac{\tan z}{\sqrt{1+\tan^2 z}}+C$$

$$=\frac{1}{4}\frac{t/2}{\sqrt{1+t^2/4}}+C=\frac{1}{4}\frac{x+1}{\sqrt{5+2x+x^2}}+C$$

16.(C) For f to be defined $x\neq-6,6$ and $\log_{0.4}\left(\frac{x-1}{x+5}\right)\geq0,\frac{x-1}{x+5}>0$. Since $\log_a x$ for $0<a<1$ is a decreasing function, we have $\frac{x-1}{x+5}\leq1$ and $\frac{x-1}{x+5}>0$. For $x>-5$, we must have $x-1\leq x+5$ and $x-1>0$. The first inequality is always true, so, we must have $x>1$. For $x<-5$, we have $x-1\geq x+5, x-1<0$. These inequalities are not possible. Hence the domain of f is $(1,\infty)\sim\{6\}$.

17.(C) $y'(x)=f'(f\left(f\left(f(x)\right)\right)f'\left(f\left(f(x)\right)\right)f'\left(f(x)\right)f'(x)$

so $y'(1)=(f'\left(f\left(f(1)\right)\right)f'\left(f\left(f(1)\right)\right)f'\left(f(1)\right)f'(1)$

$$=f'\left(f\left(f(1)\right)\right)f'\left(f(1)\right)f'(1)f'(1)=\left(f'(1)\right)^4=3^4=81$$

18.(A) In Statment-2, let **r** be the position vector of the point on the locus, then

$$|\vec{r}-\vec{a}|=|\vec{r}-\vec{b}|$$

$$\Rightarrow \quad |\vec{r}-\vec{a}|^2=|\vec{r}-\vec{b}|^2$$

$$\Rightarrow \quad |\vec{r}|^2+|\vec{a}|^2-2\vec{r}\cdot\vec{a}=|\vec{r}|^2+|\vec{b}|^2-2\vec{r}\cdot\vec{b} \quad \Rightarrow \quad 2\vec{r}\cdot(\vec{a}-\vec{b})+|\vec{b}|^2-|\vec{a}|^2=0$$

$$\Rightarrow \quad 2\vec{r}\cdot(\vec{a}-\vec{b})+(\vec{b}-\vec{a})\cdot(\vec{b}+\vec{a})=0 \quad \Rightarrow \quad \left(\vec{r}-\frac{\vec{a}+\vec{b}}{2}\right)\cdot(\vec{a}-\vec{b})=0$$

Showing that statement-2 is true using it for statement-1.

we get the required locus as

$$\left[\vec{r}-\frac{3\hat{i}-2j+5k+\hat{i}+2j-k}{2}\right]\cdot(3\hat{i}-2j+5k-(\hat{i}+2j-k))=0$$

$$\Rightarrow \quad [\vec{r}-(2\hat{i}+2k)]\cdot(2\hat{i}-4j+6k)]=0$$

$$\Rightarrow \quad \vec{r}\cdot(\hat{i}-2j+3k)=2\times1-0\times2+2\times3=8$$ and thus statement-1 is also true.

19.(D) $b_3>4b_2-3b_1$,

$$b_1r^2>4b_1r-3b_1$$

$$\left(r^2-4r+3\right)>0$$

$$r<1 \text{ or } r>3$$

20.(C) From the given condition,

2b = a + c and $b^2=\frac{2a^2c^2}{a^2+c^2}$

Eliminating b, we get $\left(\frac{a+c}{2}\right)^2=\frac{2a^2c^2}{a^2+c^2}$

$$\Rightarrow \quad (a^2+c^2)^2+2ac(a^2+c^2)-8a^2c^2=0 \quad \Rightarrow \quad [(a+c)^2+2ac](a-c)^2=0$$

$$\Rightarrow \quad 4\left(b^2+\frac{1}{2}ac\right)(a-c)^2=0 \quad \Rightarrow \quad a-c=0 \text{ or } b^2=-\frac{1}{2}ac$$

Note that $a=c,\Leftrightarrow a=b=c.$

and $b^2=-\frac{1}{2}ac$, if $a,b,-\frac{1}{2}c$ are in G.P. or $-\frac{1}{2}a,b,c$ are in G.P.

SECTION-2

1.(13) Given α,β are roots of the equation $375x^2-25x-2=0$

$$\therefore \quad \alpha+\beta=\frac{25}{375} \text{ and } \alpha\beta=\frac{-2}{375}$$

Now, $\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\alpha^r+\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\beta^r=(\alpha+\alpha^2+\ldots\infty)+(\beta+\beta^2+\ldots\infty)$

$$=\frac{\alpha}{1-\alpha}+\frac{\beta}{1-\beta}=\frac{\alpha(1-\beta)+\beta(1-\alpha)}{(1-\alpha)(1-\beta)}=\frac{\alpha+\beta-2\alpha\beta}{1-(\alpha+\beta)+\alpha\beta}=\frac{\frac{25}{375}-2\left(-\frac{2}{375}\right)}{1-\left(\frac{25}{375}\right)+\left(-\frac{2}{375}\right)}$$

$$=\frac{25-2(-2)}{375-25-2}=\frac{29}{348}=\frac{1}{12}$$

2.(3) Given:

$$H: x^2 - y^2 = 1, E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, a > b > 0$$

$$e_H = \sqrt{2}, e_E = \sqrt{1 - \frac{b^2}{a^2}}$$

We have, $e_E = \frac{1}{e_H} \quad \Rightarrow \quad \frac{a^2}{a^2 - b^2} = 2 \Rightarrow a^2 = 2b^2$

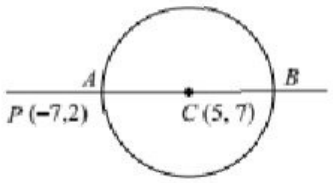
Since, $y = \sqrt{\frac{5}{2}}x + K$ is common tangent to ellipse and hyperbola, then

$$K^2 = a^2m^2 + b^2 = a^2\left(\frac{5}{2}\right) + b^2 = \frac{3}{2}$$

Put $a^2 = 2b^2$, we get $b^2 = \frac{1}{4} \therefore \quad a^2 = \frac{1}{2}$

Now, $4(a^2 + b^2) = 3$.

3.(2) Centre of the circle is C(5, 7). Let P(–7, 2) be the given point and the line joining P and C meet the circle at A and B.



Now $PC = \sqrt{(12)^2 + (5)^2} = 13$

radius of the circle is $\sqrt{5^2 + 7^2 + 51} = 5\sqrt{5}$

$p = PB = 13 + 5\sqrt{5}, q = PA = 13 - 5\sqrt{5}$

G.M of $p, q = \sqrt{pq} = \sqrt{(13)^2 - (5\sqrt{5})^2} = 2\sqrt{11} = s \quad \Rightarrow \quad \frac{s}{\sqrt{11}} = 2$

4.(1) The given equation can be written as

$$\frac{dy}{dx} + \frac{x}{1+x^2}y = \frac{x}{1+x^2}$$

This is a linear equation with I.F. $\sqrt{1+x^2}$. So

$$y\sqrt{1+x^2} = \int \frac{x}{\sqrt{1+x^2}}dx + C. = \sqrt{1+x^2} + C \quad \Rightarrow \quad y = 1 + \frac{C}{\sqrt{1+x^2}}$$

$$\frac{4}{3} = y(0) = 1 + C \Rightarrow C = \frac{1}{3}.$$

Hence $y\left(\sqrt{8}\right) = 1 + \frac{1}{9} \Rightarrow y\left(\sqrt{8}\right) - \frac{1}{9} = 1.$

5.(4) We have $\sum_{i=1}^{18}(X_i - \alpha) = 36$ and $\sum_{i=1}^{18}(X_i - \beta)^2 = 90$

$$\Rightarrow \sum_{i=1}^{18} X_i = 18(\alpha + 2) \text{ and } \sum_{i=1}^{18} X_i^2 - 2\beta\sum_{i=1}^{18} X_i + 18\beta^2 = 90 \Rightarrow \sum_{i=1}^{18} X_i^2 = 90 - 18\beta^2 + 36\beta(\alpha + 2)$$

Since, $\sigma^2 = 1$ (given)

$$\Rightarrow \frac{\sum_{i=1}^{18} X_i^2}{18} - \left(\frac{\sum_{i=1}^{18} X_i}{18}\right)^2 = 1 \qquad \Rightarrow \frac{1}{18}(90 - 18\beta^2 + 36\beta(\alpha+2)) - \left(\frac{18(\alpha+2)}{18}\right)^2 = 1$$

$$\Rightarrow 90 - 18\beta^2 + 36\beta(\alpha+2) - 18(\alpha+2)^2 = 18 \quad \Rightarrow 5 - \beta^2 + 2\alpha\beta + 4\beta - \alpha^2 - 4\alpha - 4 = 1$$

$$\Rightarrow -\alpha^2 - \beta^2 + 2\alpha\beta + 4\beta - 4\alpha = 0 \qquad \Rightarrow (\alpha-\beta)^2 + 4(\alpha-\beta) = 0$$

$$\Rightarrow (\alpha-\beta)[(\alpha-\beta)+4] = 0 \qquad \Rightarrow (\alpha-\beta) = 0 \text{ or } -4 \qquad \therefore |\alpha-\beta| = 4$$

[$\because \alpha, \beta$ are distinct real numbers]

6.(1) The given lines are $\frac{x-\lambda}{1} = \frac{y-\frac{1}{2}}{\frac{1}{2}} = \frac{z-0}{-\frac{1}{2}}$ and $\frac{x-0}{1} = \frac{y+2\lambda}{1} = \frac{z-\lambda}{1}$

The shortest distance between the lines is given by

$$\left| \frac{\begin{vmatrix} 0-\lambda & -2\lambda-\frac{1}{2} & \lambda-0 \\ 1 & \frac{1}{2} & -\frac{1}{2} \\ 1 & 1 & 1 \end{vmatrix}}{\sqrt{\left(\frac{1}{2}+\frac{1}{2}\right)^2 + \left(-\frac{1}{2}-1\right)^2 + \left(1-\frac{1}{2}\right)^2}} \right| = \frac{\sqrt{7}}{2\sqrt{2}}$$

$$\Rightarrow \left| \frac{-\lambda\left(\frac{1}{2}+\frac{1}{2}\right) + \left(2\lambda+\frac{1}{2}\right)\left(1+\frac{1}{2}\right) + \lambda\left(1-\frac{1}{2}\right)}{\frac{\sqrt{14}}{2}} \right| = \frac{\sqrt{7}}{2\sqrt{2}}$$

$$\Rightarrow \left| -\lambda + 3\lambda + \frac{3}{4} + \frac{\lambda}{2} \right| = \frac{7}{4} \qquad \Rightarrow \left| \frac{5\lambda}{2} + \frac{3}{4} \right| = \frac{7}{4} \Rightarrow \frac{5\lambda}{2} + \frac{3}{4} = \pm\frac{7}{4}$$

$$\Rightarrow \text{Either } \Rightarrow \frac{5\lambda}{2} = \frac{7}{4} - \frac{3}{4} \text{ or } \frac{5\lambda}{2} = -\frac{7}{4} - \frac{3}{4} \qquad \Rightarrow \frac{5\lambda}{2} = 1 \text{ or } \frac{5\lambda}{2} = \frac{-10}{4}$$

$$\Rightarrow \lambda = \frac{2}{5} \text{ or } \lambda = -1 \Rightarrow \lambda = -1 \qquad [\because \lambda \text{ is an integer}]$$

$$\therefore \quad |\lambda| = 1$$

7.(8) We have,

$$(2x^2+3x+4)^{10}=\sum_{r=0}^{20}a_r x^r \qquad \ldots(i)$$

Replacing x by $\frac{2}{x}$, we get

$$\left(\frac{8}{x^2}+\frac{6}{x}+4\right)^{10}=\sum_{r=0}^{20}a_r\left(\frac{2}{x}\right)^r \Rightarrow \frac{2^{10}(2x^2+3x+4)^{10}}{x^{20}}=\sum_{r=0}^{20}a_r 2^r x^{-r}$$

$$\Rightarrow \sum_{r=0}^{20}a_r x^r=\sum_{r=0}^{20}a_r 2^{r-10}x^{20-r} \qquad \text{[Using (i)]}$$

On comparing the coefficient of x^7 from both sides, we get, $a_7=a_{13}2^3 \Rightarrow \frac{a_7}{a_{13}}=8$

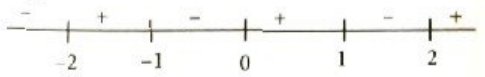
8.(5) We have,

$$f(x)=\int_0^{x^2}\frac{t^2-5t+4}{2+e^t}dt$$

$$f'(x)=\frac{(x^4-5x^2+4)\cdot 2x}{2+e^{x^2}}=\frac{2x(x^2-4)(x^2-1)}{(2+e^{x^2})}$$

For critical points, $2x(x^2-4)(x^2-1)=0 \quad \Rightarrow x=0,\pm 2,\pm 1.$

Sign scheme of f '(x) is



Hence, $m=2$, $n=3$

9.(12) Given $y+z=5$ …(i)

and $\frac{1}{y}+\frac{1}{z}=\frac{5}{6}, y>z$ …(ii)

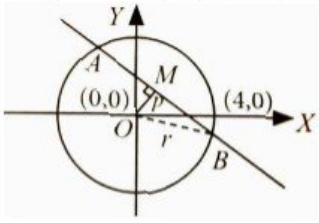
Solving (i) and (ii), we get $y=3$, $z=2$

$\Rightarrow \quad n=2^x\cdot 3^3\cdot 5^2=(2\cdot 2\cdot 2....)(3\cdot 3\cdot 3)(5\cdot 5)$

$\therefore$ Number of odd divisors = (3 + 1)(2 + 1) = 4 × 3 = 12

10.(210)

Let p be the perpendicular distance from O(0, 0) to line $x+y=n$.



$\therefore \quad p=\frac{n}{\sqrt{2}}<4$ for n = 1, 2, 3, 4, 5

Now, length of chord $=2\sqrt{r^2-p^2}=2\sqrt{r^2-\frac{n^2}{2}}=2\sqrt{16-\frac{n^2}{2}}=\sqrt{64-2n^2}$

Now, sum of the squares of the chords = 62 + 56 + 46 + 32 + 14 = 210


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